ADVANCED SQL QUERY DOCUMENT

# NAILVERSE

*Not just NAILS.. IT's the WHOLE MOOD*

*Online Nail Art & Nail Beauty Online Platform*

***WEEK - 10***

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## 10. Advanced SQL Query System

### Scenario

Nailverse now wants answers to harder questions than a simple SELECT can give. Which services are priced above the usual level? Which customers spend more than the average customer? Which category earns the most, and which service does each customer book most often? Each of these questions needs the result of one query to be used inside another.

This week uses subqueries and nested queries on the existing Nailverse tables to find premium services, high-value customers and business trends, and finishes with four advanced business reports.

### 1. Implement Subqueries and Nested Queries

**Concept:**

A subquery is a query written inside another SQL query. The inner query is solved first and its result is used by the outer query. This makes it possible to answer a question that depends on a calculated value, such as an average or a maximum.

**Types covered:**

- Single-row subquery – the inner query returns one value (for example, the average price).
- Multi-row subquery – the inner query returns a list of values, used with IN or ANY.
- Nested subquery – a subquery placed inside another subquery.

**Example: find the services whose price is greater than the average service price.**

Step 1: Find the average price.

```
SQL> SELECT ROUND(AVG(Price), 2) AS Average_Price
  2  FROM Service;

AVERAGE_PRICE
-------------
       784.17
```

Step 2: Compare each service with the average price.

```
SQL> SELECT Service_Name, Price
  2  FROM Service
  3  WHERE Price >
  4  (SELECT AVG(Price) FROM Service);

SERVICE_NAME                                                PRICE
---------------------------------------------------------- ----------
CHROME NAIL ART                                               999
SPA PEDICURE                                                  799
Chrome Nail Art                                              1299
```

## 2. Find Services Above Average Price

Task: The salon wants to identify its premium services:

1. Find the services priced above the average price.

```
SQL> SELECT Service_ID, Service_Name, Price
  2  FROM Service
  3  WHERE Price > (SELECT AVG(Price) FROM Service);

SERVICE SERVICE_NAME                                        PRICE
------- ---------------------------------------------- ----------
SRV004  CHROME NAIL ART                                       999
SRV005  SPA PEDICURE                                          799
SRV006  Chrome Nail Art                                      1299
```

2. Display the service name, category and price.

```
SQL> SELECT S.Service_Name AS Service, S.Price, C.Category_Name AS Category
  2  FROM Service S
  3  JOIN Category C ON S.Category_ID = C.Category_ID
  4  WHERE S.Price > (SELECT AVG(Price) FROM Service);

SERVICE                                                    PRICE
---------------------------------------------------------- ----------
CATEGORY
------------------------------
CHROME NAIL ART                                               999
NAIL ART

SPA PEDICURE                                                  799
SPA CARE

Chrome Nail Art                                              1299
NAIL EXTENSIONS
```

3. Identify the expensive services in each category (price above the average of its own category).

```
SQL> SELECT c.Category_Name AS Category,
  2         s.Service_Name AS Service,
  3         s.Price
  4  FROM Service s
  5  JOIN Category c ON s.Category_ID = c.Category_ID
  6  WHERE s.Price > (
  7      SELECT AVG(s2.Price)
  8      FROM Service s2
  9      WHERE s2.Category_ID = s.Category_ID
 10  )
 11  ORDER BY c.Category_Name, s.Price;

CATEGORY
------------------------------
SERVICE                                                 PRICE
-------------------------------------------------- ----------
NAIL EXTENSIONS
Chrome Nail Art                                          1299
```

This is a correlated subquery: the inner query is run again for every service, using that service's own category.

4. Find the services priced higher than the average of a selected category (Manicure).

```
SQL> SELECT Service_Name, Price
  2  FROM Service
  3  WHERE Price > (
  4      SELECT AVG(s2.Price)
  5      FROM Service s2
  6      JOIN Category c2 ON s2.Category_ID = c2.Category_ID
  7      WHERE c2.Category_Name = 'MANICURE'
  8  )
  9  ORDER BY Price;

SERVICE_NAME                                            PRICE
-------------------------------------------------- ----------
CLASSIC PEDICURE                                           650
SPA PEDICURE                                               799
CHROME NAIL ART                                            999
Chrome Nail Art                                           1299
```

## 3. Identify Customers with Maximum Purchases

Task: Analyze customer booking behavior. Find:

1. The customer who spent the highest amount.

```
SQL> SELECT Customer_ID,
  2         SUM(Total_Amount) AS Total_Spending
  3  FROM Appointment
  4  GROUP BY Customer_ID
  5  HAVING SUM(Total_Amount) = (
  6      SELECT MAX(Total_Sales)
  7      FROM (
  8          SELECT SUM(Total_Amount) AS Total_Sales
  9          FROM Appointment
 10          GROUP BY Customer_ID
 11      )
 12  );

CUSTOMER_ID          TOTAL_SPENDING
-------------------- --------------
CU-1006                        1299
```

2. The customers with the maximum number of appointments.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         COUNT(a.Appointment_ID) AS Number_Of_Appointments
  4  FROM Customer c
  5  JOIN Appointment a ON c.Customer_ID = a.Customer_ID
  6  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
  7  HAVING COUNT(a.Appointment_ID) = (
  8      SELECT MAX(Appointment_Count)
  9      FROM (
 10          SELECT COUNT(*) AS Appointment_Count
 11          FROM Appointment
 12          GROUP BY Customer_ID
 13      )
 14  );

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
NUMBER_OF_APPOINTMENTS
----------------------
CU-1004              PRIYA NAIR
                     1

CU-1006              VARSHA RAJAN
                     1

CU-1007              MOSHAMI JUN
                     1


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
NUMBER_OF_APPOINTMENTS
----------------------
CU-1009              SRI VALLI
                     1

CU-1002              KAVYA RAO
                     1

CU-1010              NILLA VEERA
                     1
```

```
CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
NUMBER_OF_APPOINTMENTS
----------------------
CU-1005              NISHA KUMARI
                     1

CU-1003              ANANYA  SURESH
                     1

CU-1008              HARI KUMAR
                     1


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
NUMBER_OF_APPOINTMENTS
----------------------
CU-1001              MEERA IYER
                     1


10 rows selected.
```

3.The customers whose spending is above the average customer spending.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         SUM(a.Total_Amount) AS Total_Spending
  4  FROM Customer c
  5  JOIN Appointment a ON c.Customer_ID = a.Customer_ID
  6  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
  7  HAVING SUM(a.Total_Amount) > (
  8      SELECT AVG(Customer_Total)
  9      FROM (
 10          SELECT SUM(Total_Amount) AS Customer_Total
 11          FROM Appointment
 12          GROUP BY Customer_ID
 13      )
 14  )
 15  ORDER BY Total_Spending;

CUSTOMER_ID          CUSTOMER_NAME                                TOTAL_SPENDING
-------------------- ---------------------------------------- --------------
CU-1005              NISHA KUMARI                                            799
CU-1004              PRIYA NAIR                                              999
CU-1010              NILLA VEERA                                             999
CU-1006              VARSHA RAJAN                                           1299
```

4. The top 5 valuable customers.

```
SQL> SELECT *
  2  FROM (
  3      SELECT c.Customer_ID,
  4             c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  5             SUM(a.Total_Amount) AS Total_Spending
  6      FROM Customer c
  7      JOIN Appointment a ON c.Customer_ID = a.Customer_ID
  8      GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
  9      ORDER BY Total_Spending DESC
 10  )
 11  WHERE ROWNUM <= 5;

CUSTOMER_ID          CUSTOMER_NAME                            TOTAL_SPENDING
-------------------- ---------------------------------------- --------------
CU-1006              VARSHA RAJAN                                       1299
CU-1004              PRIYA NAIR                                          999
CU-1010              NILLA VEERA                                         999
CU-1005              NISHA KUMARI                                        799
CU-1002              KAVYA RAO                                           650
```

## 4. Generate Complex Business Queries

### Query 1: Best-Selling Service

Find the services with the maximum sales quantity. Display the service name, total quantity sold and total revenue generated.

```
SQL> SELECT s.Service_Name,
  2         SUM(ad.Quantity) AS Total_Quantity_Sold,
  3         SUM(ad.Quantity * ad.Price) AS Total_Revenue
  4  FROM Service s
  5  JOIN Appointment_Details ad ON s.Service_ID = ad.Service_ID
  6  GROUP BY s.Service_ID, s.Service_Name
  7  HAVING SUM(ad.Quantity) = (
  8      SELECT MAX(Qty)
  9      FROM (
 10          SELECT SUM(Quantity) AS Qty
 11          FROM Appointment_Details
 12          GROUP BY Service_ID
 13      )
 14  );

SERVICE_NAME                                       TOTAL_QUANTITY_SOLD
-------------------------------------------------- -------------------
TOTAL_REVENUE
-------------
NAIL EXTENSIONS                                                      2
          918

CLASSIC MAINCURE                                                     2
          998

CHROME NAIL ART                                                      2
         1998


SERVICE_NAME                                       TOTAL_QUANTITY_SOLD
-------------------------------------------------- -------------------
TOTAL_REVENUE
-------------
CLASSIC PEDICURE                                                     2
         1300
```

### Query 2: High-Value Customers

Find the customers whose total booking amount is greater than the average customer spending.

```
SQL> SELECT c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  2         SUM(a.Total_Amount) AS Total_Spending,
  3         ROUND((SELECT AVG(t.Customer_Total)
  4                FROM (
  5                    SELECT SUM(Total_Amount) AS Customer_Total
  6                    FROM Appointment
  7                    GROUP BY Customer_ID
  8                ) t), 2) AS Average_Spending
  9  FROM Customer c
 10  JOIN Appointment a ON c.Customer_ID = a.Customer_ID
 11  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
 12  HAVING SUM(a.Total_Amount) > (
 13      SELECT AVG(t2.Customer_Total)
 14      FROM (
 15          SELECT SUM(Total_Amount) AS Customer_Total
 16          FROM Appointment
 17          GROUP BY Customer_ID
 18      ) t2
 19  )
 20  ORDER BY Total_Spending DESC;

CUSTOMER_NAME                          TOTAL_SPENDING AVERAGE_SPENDING
-------------------------------------- -------------- ----------------
VARSHA RAJAN                                     1299            731.2
NILLA VEERA                                       999            731.2
PRIYA NAIR                                        999            731.2
NISHA KUMARI                                      799            731.2
```

### Query 3: Category Performance Analysis

Find the highest revenue-generating category, the number of services sold in each category and the average service price by category.

- Highest revenue-generating category

```
SQL> SELECT c.Category_Name,
  2         SUM(ad.Quantity * ad.Price) AS Total_Revenue
  3  FROM Category c
  4  JOIN Service s ON c.Category_ID = s.Category_ID
  5  JOIN Appointment_Details ad ON s.Service_ID = ad.Service_ID
  6  GROUP BY c.Category_ID, c.Category_Name
  7  HAVING SUM(ad.Quantity * ad.Price) = (
  8      SELECT MAX(Category_Revenue)
  9      FROM (
 10          SELECT SUM(ad2.Quantity * ad2.Price) AS Category_Revenue
 11          FROM Service s2
 12          JOIN Appointment_Details ad2 ON s2.Service_ID = ad2.Service_ID
 13          GROUP BY s2.Category_ID
 14      )
 15  );

CATEGORY_NAME                 TOTAL_REVENUE
----------------------------- -------------
NAIL EXTENSIONS                        2217
```

- Number of services sold and average service price in each category

```
SQL> SELECT c.Category_Name,
  2         (SELECT NVL(SUM(ad.Quantity), 0)
  3          FROM Service s
  4          JOIN Appointment_Details ad ON s.Service_ID = ad.Service_ID
  5          WHERE s.Category_ID = c.Category_ID) AS Services_Sold,
  6         (SELECT ROUND(AVG(s2.Price), 2)
  7          FROM Service s2
  8          WHERE s2.Category_ID = c.Category_ID) AS Average_Price
  9  FROM Category c
 10  ORDER BY Services_Sold DESC;

CATEGORY_NAME                 SERVICES_SOLD AVERAGE_PRICE
----------------------------- ------------- -------------
NAIL EXTENSIONS                           3           879
PEDICURE                                  2           650
NAIL ART                                  2           999
MANICURE                                  2           499
SPA CARE                                  1           799

SQL>
```

### Query 4: Customer Booking History

Display the customer name, number of appointments, total booking amount and the most booked service. If two services are tied, the first one alphabetically is shown.

```
SQL> WITH Cust_Service AS (
  2      SELECT a.Customer_ID,
  3             ad.Service_ID,
  4             SUM(ad.Quantity) AS Qty
  5      FROM Appointment a
  6      JOIN Appointment_Details ad
  7        ON a.Appointment_ID = ad.Appointment_ID
  8      GROUP BY a.Customer_ID, ad.Service_ID
  9  )
 10  SELECT c.First_Name || ' ' || c.Last_Name AS Customer_Name,
 11         (SELECT COUNT(*)
 12          FROM Appointment a
 13          WHERE a.Customer_ID = c.Customer_ID) AS Number_Of_Appointments,
 14         (SELECT NVL(SUM(a.Total_Amount), 0)
 15          FROM Appointment a
 16          WHERE a.Customer_ID = c.Customer_ID) AS Total_Amount,
 17         (SELECT MIN(s.Service_Name)
 18          FROM Cust_Service cs
 19          JOIN Service s ON cs.Service_ID = s.Service_ID
 20          WHERE cs.Customer_ID = c.Customer_ID
 21            AND cs.Qty = (
 22                SELECT MAX(cs2.Qty)
 23                FROM Cust_Service cs2
 24                WHERE cs2.Customer_ID = c.Customer_ID
 25            )) AS Most_Booked_Service
 26  FROM Customer c
 27  ORDER BY Total_Amount DESC, Customer_Name;

CUSTOMER_NAME                              NUMBER_OF_APPOINTMENTS TOTAL_AMOUNT
------------------------------------------ ---------------------- ------------
MOST_BOOKED_SERVICE
--------------------------------------------------

VARSHA RAJAN                                                    1         1299
Chrome Nail Art

NILLA VEERA                                                     1          999
CHROME NAIL ART

PRIYA NAIR                                                      1          999
CHROME NAIL ART


CUSTOMER_NAME                              NUMBER_OF_APPOINTMENTS TOTAL_AMOUNT
------------------------------------------ ---------------------- ------------
MOST_BOOKED_SERVICE
--------------------------------------------------

NISHA KUMARI                                                    1          799
SPA PEDICURE

HARI KUMAR                                                      1          650
CLASSIC PEDICURE

KAVYA RAO                                                       1          650
CLASSIC PEDICURE
```

```
CUSTOMER_NAME                              NUMBER_OF_APPOINTMENTS TOTAL_AMOUNT
------------------------------------------ ---------------------- ------------
MOST_BOOKED_SERVICE
--------------------------------------------------

MEERA IYER                                                      1          499
CLASSIC MAINCURE

MOSHAMI JUN                                                     1          499
CLASSIC MAINCURE

ANANYA  SURESH                                                  1          459
NAIL EXTENSIONS


CUSTOMER_NAME                              NUMBER_OF_APPOINTMENTS TOTAL_AMOUNT
------------------------------------------ ---------------------- ------------
MOST_BOOKED_SERVICE
--------------------------------------------------

SRI VALLI                                                       1          459
NAIL EXTENSIONS


10 rows selected.
```

## 5. Prepare Advanced SQL Report

### Report 1: Premium Service Report

Display the service name, category, price and slot availability. Condition: services priced above the average price.

```
SQL> SELECT s.Service_Name,
  2         c.Category_Name AS Category,
  3         s.Price
  4  FROM Service s
  5  JOIN Category c ON s.Category_ID = c.Category_ID
  6  WHERE s.Price > (SELECT AVG(Price) FROM Service)
  7  ORDER BY s.Price DESC;

SERVICE_NAME
--------------------------------------------------
CATEGORY                          PRICE
------------------------------ ----------
Chrome Nail Art
NAIL EXTENSIONS                     1299

CHROME NAIL ART
NAIL ART                             999

SPA PEDICURE
SPA CARE                             799
```

### Report 2: Customer Value Report

Display the customer name, total appointments, total spending and customer category. A customer is Premium if the spending is above the average customer spending, Regular if the customer has booked but spent less, and New if there is no booking yet.

```
SQL> SELECT c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  2         COUNT(a.Appointment_ID) AS Total_Appointments,
  3         NVL(SUM(a.Total_Amount), 0) AS Total_Spending,
  4         CASE
  5           WHEN NVL(SUM(a.Total_Amount), 0) > (
  6               SELECT AVG(t.Customer_Total)
  7               FROM (
  8                   SELECT SUM(Total_Amount) AS Customer_Total
  9                   FROM Appointment
 10                   GROUP BY Customer_ID
 11               ) t
 12           ) THEN 'Premium'
 13           WHEN COUNT(a.Appointment_ID) > 0 THEN 'Regular'
 14           ELSE 'New'
 15         END AS Customer_Category
 16  FROM Customer c
 17  LEFT JOIN Appointment a ON c.Customer_ID = a.Customer_ID
 18  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
 19  ORDER BY Total_Spending DESC;

CUSTOMER_NAME                          TOTAL_APPOINTMENTS TOTAL_SPENDING
-------------------------------------- ------------------ --------------
CUSTOME
-------
VARSHA RAJAN                                            1           1299
Premium

PRIYA NAIR                                              1            999
Premium

NILLA VEERA                                             1            999
Premium


CUSTOMER_NAME                          TOTAL_APPOINTMENTS TOTAL_SPENDING
-------------------------------------- ------------------ --------------
CUSTOME
-------
NISHA KUMARI                                            1            799
Premium

KAVYA RAO                                               1            650
Regular

HARI KUMAR                                              1            650
Regular
```

```
CUSTOMER_NAME                          TOTAL_APPOINTMENTS TOTAL_SPENDING
-------------------------------------- ------------------ --------------
CUSTOME
-------
MOSHAMI JUN                                             1            499
Regular

MEERA IYER                                              1            499
Regular

ANANYA  SURESH                                          1            459
Regular


CUSTOMER_NAME                          TOTAL_APPOINTMENTS TOTAL_SPENDING
-------------------------------------- ------------------ --------------
CUSTOME
-------
SRI VALLI                                               1            459
Regular


10 rows selected.
```

### Report 3: Sales Performance Report

- Service-wise sales

```
SQL> SELECT s.Service_Name,
  2         NVL(SUM(ad.Quantity), 0) AS Quantity_Sold,
  3         NVL(SUM(ad.Quantity * ad.Price), 0) AS Revenue
  4  FROM Service s
  5  LEFT JOIN Appointment_Details ad
  6    ON s.Service_ID = ad.Service_ID
  7  GROUP BY s.Service_ID, s.Service_Name
  8  ORDER BY Revenue DESC, s.Service_Name;

SERVICE_NAME                                       QUANTITY_SOLD    REVENUE
-------------------------------------------------- ------------- ----------
CHROME NAIL ART                                                2       1998
CLASSIC PEDICURE                                               2       1300
Chrome Nail Art                                                1       1299
CLASSIC MAINCURE                                               2        998
NAIL EXTENSIONS                                                2        918
SPA PEDICURE                                                   1        799

6 rows selected.
```

- Category-wise revenue

```
SQL> SELECT c.Category_Name,
  2         NVL(SUM(ad.Quantity * ad.Price), 0) AS Revenue
  3  FROM Category c
  4  JOIN Service s ON c.Category_ID = s.Category_ID
  5  LEFT JOIN Appointment_Details ad ON s.Service_ID = ad.Service_ID
  6  GROUP BY c.Category_ID, c.Category_Name
  7  ORDER BY Revenue DESC;

CATEGORY_NAME                    REVENUE
------------------------------ ----------
NAIL EXTENSIONS                      2217
NAIL ART                             1998
PEDICURE                             1300
MANICURE                              998
SPA CARE                              799
```

- Top-selling services (booked 3 or more times)\

```
SQL> SELECT s.Service_Name,
  2         SUM(ad.Quantity) AS Quantity_Sold
  3  FROM Service s
  4  JOIN Appointment_Details ad ON s.Service_ID = ad.Service_ID
  5  GROUP BY s.Service_ID, s.Service_Name
  6  HAVING SUM(ad.Quantity) >= 2
  7  ORDER BY Quantity_Sold DESC, s.Service_Name;

SERVICE_NAME                                       QUANTITY_SOLD
-------------------------------------------------- -------------
CHROME NAIL ART                                                2
CLASSIC MAINCURE                                               2
CLASSIC PEDICURE                                               2
NAIL EXTENSIONS                                                2
```

**Report 4: Business Decision Report**

- High-performing services (revenue above the average service revenue)

```
SQL> SELECT s.Service_Name,
  2         SUM(ad.Quantity * ad.Price) AS Revenue
  3  FROM Service s
  4  JOIN Appointment_Details ad ON s.Service_ID = ad.Service_ID
  5  GROUP BY s.Service_ID, s.Service_Name
  6  HAVING SUM(ad.Quantity * ad.Price) > (
  7      SELECT AVG(Service_Revenue)
  8      FROM (
  9          SELECT SUM(Quantity * Price) AS Service_Revenue
 10          FROM Appointment_Details
 11          GROUP BY Service_ID
 12      )
 13  )
 14  ORDER BY Revenue DESC;

SERVICE_NAME                                       REVENUE
-------------------------------------------------- ----------
CHROME NAIL ART                                          1998
CLASSIC PEDICURE                                         1300
Chrome Nail Art                                          1299
```

- Low-performing services (revenue below average, including services never booked)

```
SQL> SELECT s.Service_Name,
  2         NVL(SUM(ad.Quantity * ad.Price), 0) AS Revenue
  3  FROM Service s
  4  LEFT JOIN Appointment_Details ad
  5  ON s.Service_ID = ad.Service_ID
  6  GROUP BY s.Service_ID, s.Service_Name
  7  HAVING NVL(SUM(ad.Quantity * ad.Price), 0) < (
  8      SELECT AVG(Service_Revenue)
  9      FROM (
 10          SELECT SUM(Quantity * Price) AS Service_Revenue
 11          FROM Appointment_Details
 12          GROUP BY Service_ID
 13      )
 14  )
 15  ORDER BY Revenue, s.Service_Name;

SERVICE_NAME                                       REVENUE
-------------------------------------------------- ----------
SPA PEDICURE                                             799
NAIL EXTENSIONS                                          918
CLASSIC MAINCURE                                         998
```

- High-value customers

```
SQL> SELECT c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  2         SUM(a.Total_Amount) AS Total_Spending
  3  FROM Customer c
  4  JOIN Appointment a ON c.Customer_ID = a.Customer_ID
  5  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
  6  HAVING SUM(a.Total_Amount) > (
  7      SELECT AVG(t.Customer_Total)
  8      FROM (
  9          SELECT SUM(Total_Amount) AS Customer_Total
 10          FROM Appointment
 11          GROUP BY Customer_ID
 12      ) t
 13  )
 14  ORDER BY Total_Spending DESC;

CUSTOMER_NAME                             TOTAL_SPENDING
----------------------------------------- --------------
VARSHA RAJAN                                        1299
NILLA VEERA                                          999
PRIYA NAIR                                           999
NISHA KUMARI                                         799
```

- Revenue trend (week by week)

```
SQL> SELECT CASE
  2           WHEN Appointment_Date <= DATE '2026-09-07'
  3             THEN 'Week 1 (1-7 Sep)'
  4           WHEN Appointment_Date <= DATE '2026-09-14'
  5             THEN 'Week 2 (8-14 Sep)'
  6           ELSE 'Week 3 (15-21 Sep)'
  7         END AS Booking_Week,
  8         COUNT(*) AS Appointments,
  9         SUM(Total_Amount) AS Revenue
 10  FROM Appointment
 11  GROUP BY CASE
 12             WHEN Appointment_Date <= DATE '2026-09-07'
 13               THEN 'Week 1 (1-7 Sep)'
 14             WHEN Appointment_Date <= DATE '2026-09-14'
 15               THEN 'Week 2 (8-14 Sep)'
 16             ELSE 'Week 3 (15-21 Sep)'
 17           END
 18  ORDER BY MIN(Appointment_Date);

BOOKING_WEEK       APPOINTMENTS    REVENUE
------------------ ------------ ----------
Week 1 (1-7 Sep)              5       3406
Week 3 (15-21 Sep)            5       3906
```

## Conclusion

Subqueries and nested queries made it possible to compare every service and customer with an average or a maximum calculated from the same data. The reports identify the premium services, the customers who spend above average, the best and weakest performing services and categories, and how revenue moved week by week. Together they give Nailverse the information needed for pricing, promotion and customer-retention decisions.